Inha University in Tashkent
School of Computer Science and Information Engineering
Course Name: Object Oriented Programming 2
Semester and Year: Spring Freshman, 2026
Laboratory Assignment 5: Object-Oriented Programming in C++

Note: 1. Submission only via E-CLASS is acceptable (If you have any issue with E-CLASS, Please consult IT Dept.)

2. Comments should be included in your source code.

3. Do not plagiarize or use AI tools to complete the assignment. If a student is caught doing so, he/she will receive ZERO for the assignment.

4. Upload your completed laboratory exercise as a zip file including the appropriate files **(.cpp, and .h),** to the eClass system before the deadline. The zip file must be named with your **ID number.**

Exercise 5: University Management System

Objective: Design a C++ program that demonstrates inheritance by modeling different people in a university system.

Tasks:

1. **Create Base class:**
    - Define a class Person.
    - Declare the following protected members: Name (string), age (int) and id (int).
    - Define parameterized constructor.
    - Define displayPerson() to print the person's details.
    - Define setPerson() and validate that age and ID are positive values.
2. **Create Derived Class:**
    - Define a class Student (public inheritance).
    - Declare the following private members for Student class: major (string),

gpa(double) and semester (int).

- Define parameterized constructor. Constructor of Student must call constructor of Person.
- Define displayStudent() to print the student's details.
- Define setStudent() and validate that GPA must be between 0.0 and 4.0, and semester must be positive..

3. **Create Derived Class:**
   - Define a class Teacher (public inheritance).
   - Declare the following private members for Teacher class: department (string), salary (double) and subject (string).
   - Define parameterized constructor. Constructor of Teacher must call constructor of Person.
   - Define displayTeacher() to print the teacher's details.
   - Define setTeacher() and validate that salary is positive value.
4. **Create Derived Class:**
   - Define a class Staff (public inheritance).
   - Declare the following private members for Staff class: position (string), monthlySalary (double) and workingHours (int).
   - Define parameterized constructor. Constructor of Staff must call constructor of Person.
   - Define displayStaff() to print the staff's details.
   - Define setStaff() and validate that monthlySalary and workingHours are positive values.

In main():

- Create at least 2 Student objects, 2 Teacher objects and 2 Staff objects. For each object initialize values using constructors and display full details.

Output:

```
----- Student Details -----
Name: Ali
Age: 20
ID: 101
Major: Computer Science
GPA: 3.8
Semester: 3

----- Student Details -----
Name: Sara
Age: 19
ID: 102
Major: Software Engineering
GPA: 3.5
Semester: 2

----- Teacher Details -----
Name: Mr. Karimov
Age: 40
ID: 201
Department: Engineering
Salary: 5000
Subject: C++ Programming

----- Teacher Details -----
Name: Ms. Nargiza
Age: 35
ID: 202
Department: Mathematics
Salary: 4500
Subject: Discrete Math

----- Staff Details -----
Name: Jasur
Age: 30
ID: 301
Position: Administrator
Monthly Salary: 3000
Working Hours: 8

----- Staff Details -----
Name: Dilshod
Age: 45
```